

Morphological Constraints on Intracohort Cannibalism in Age-0 Largemouth Bass

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Abstract.—Intracohort cannibalism can be an important process shaping the growth and survival of age-0 largemouth bass *Micropterus salmoides*. To assess the morphological constraints on intracohort cannibalism in largemouth bass, we evaluated the relationship of mouth width, body depth, and total length in age-0 fish. Morphological measurements predicted a threshold predator–prey relationship of: prey length = $0.309 \times$ predator length^{1.109}, whereas experimental results suggested the relationship of: prey length = $0.5 \times$ predator length. Our results indicated that intracohort cannibalism may begin as early as 9 d after swim-up in southern lakes and reservoirs and 12 d in northern lakes. However, because of the length of the spawning season, slower growth, and male guarding, intracohort cannibalism may be more important in southern systems than in northern lakes.

Cannibalism is an important process regulating population dynamics (Fox 1975; Polis 1981) and is prevalent among many fish species (Smith and Reay 1991). Largemouth bass *Micropterus salmoides* exhibit inter- and intracohort cannibalism in the early stages of life (Cooper 1937; DeAngelis et al. 1979). During the first year of life, intracohort cannibalism may allow early spawned or fast-growing individuals to gain a substantial growth advantage by capitalizing on a valuable prey resource (Kitchell et al. 1994; Brabrand 1995). As a result, intracohort cannibalism may constitute a substantial source of mortality for later-spawned or slow-growing individuals (DeAngelis et al. 1979; Brabrand 1995). Therefore, the size at which intracohort cannibalism first occurs and the relationship between prey and predator size are fundamental to understanding the effect of intracohort cannibalism on population dynamics.

For piscivorous fish, mouth width (gape) relative to prey body depth determines whether prey can be ingested (Lawrence 1958; Werner 1974; Webb 1986; Hambricht 1991). This relationship of prey body depth to predator mouth width sets an upper

limit on the maximum prey size available to piscivorous fish. Lawrence (1958) examined body depth and mouth width of largemouth bass using morphometric measurements. His work, however, included few age-0 bass and none less than 35 mm. Extrapolation of Lawrence's (1958) work to small age-0 largemouth bass suggested that 10-mm largemouth bass could ingest a 12-mm largemouth bass. Because of the lack of information on body depth and mouth width for age-0 largemouth bass, our study focused on cannibalism among largemouth bass during their first year of life.

Here we present predator–prey relationships for age-0 largemouth bass developed from morphometric measurements and a series of predator–prey microcosm experiments. Our goals were to quantify the size of first potential intracohort cannibalism in largemouth bass and to quantify the predator:prey length ratio for potential cannibalism throughout the first year of life.

Methods

Age-0 largemouth bass were collected by dip net and beach seine during summer 1994 from Paul, Bay, and Fish lakes. Paul and Bay lakes are at the University of Notre Dame Environmental Research Center in southern Gogebic County, Michigan. Fish Lake is in northwest Dane County, Wisconsin.

We measured total length (TL), body depth, and mouth width of 220 fish ranging from about 7 to 100 mm long. Total length was measured to the nearest 1 mm. Body depth and mouth width were measured to the nearest 0.18 or 0.084 mm with a dissecting microscope. Body depths of individuals exceeding 80 mm TL were measured to the nearest 1 mm. Body depths were measured from the anterior edge of the dorsal fin to the bottom of the abdomen, tangent to the edge of the opercular flap. Mouth width was measured as the distance between the outer edges of the maxillary bones just beneath the eyes with the mouth closed. Lawrence (1956) showed that this measure of mouth width was a reliable metric of gape size.

The relationship of total length and body depth,

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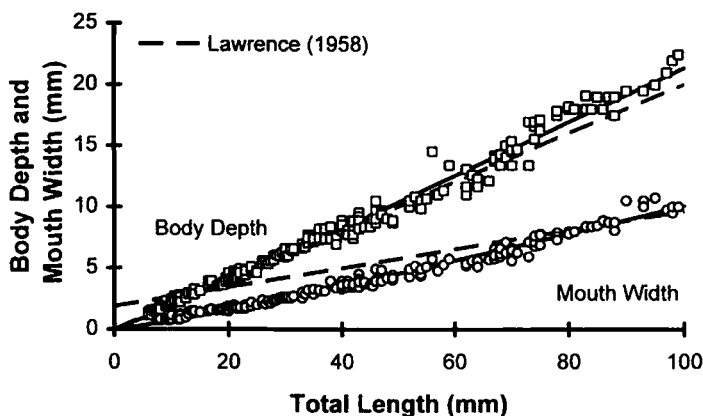


FIGURE 1.—Relationships for total length versus body depth and total length versus mouth width for age-0 largemouth bass from the present study (solid line). Dashed lines are predictions based on Lawrence (1958).

and total length and mouth width were estimated with the model $Y = aX^b$, where Y was body depth or mouth width and X was total length. Model parameters were estimated by least squares, and residuals of each model were checked for normality (Draper and Smith 1981).

Seventy-three predator-prey feeding trials were run in 57–76-L aquaria. Water temperature was maintained between 15 and 20°C for all trials. In each trial, we presented 5–10 prey of a single size (± 1 –2 mm) to one predator. Predators were measured to the nearest 1 mm before each trial. Similar sized prey were subsampled from nest schools. The nest school was then subsampled again to determine the size range of prey presented to each predator. Predator-prey combinations were chosen to bracket the expected predator:prey ratio. Each predator and prey were used in only one trial to retain statistical independence. Before trials began, predators were acclimated to experimental aquaria for 6 h and were not fed during this period. Prey were then added to aquaria. After 24 h, each aquarium was emptied and the number of prey eaten was determined. We considered cannibalism possible for a given predator-prey combination if one prey was eaten. In trials in which cannibalism occurred, an average of 70% of the presented prey were eaten.

The size of first potential cannibalism was calculated from morphological relationships and experimental results. We assumed a swim-up size of 6 mm for all fish (Brown 1985; Goodgame and Miranda 1993). The size of first potential cannibalism was the minimum size a predator had to be to ingest a 6-mm prey as calculated from mor-

phological measurements and as observed in experimental trials.

Results

The relationships between total length and mouth width, and total length and body depth for age-0 largemouth bass (Figure 1) were

$$\text{mouth width} = 0.0507 \cdot \text{length}^{1.149}$$

($R^2 = 0.994$), and

$$\text{body depth} = 0.173 \cdot \text{length}^{1.046}$$

($R^2 = 0.995$). When mouth width and body depth were equated, the relationship between predator and prey length for age-0 largemouth bass cannibalism was

$$\text{prey length} = 0.309 \cdot \text{predator length}^{1.099}.$$

This equation represents the maximum length of largemouth bass that may be ingested by an age-0 largemouth bass of a given size (predator:prey ratio < 2:1).

Our laboratory feeding trials, however, indicated that predators were able to ingest fish larger than those predicted by our morphological measurements. The realized maximum possible length of prey may be closer to 50% of the predator's body length (Figure 2).

Largemouth bass may become cannibalistic at a small size. Morphological measurements (from the equation describing the relation of prey length to predator length) predict that cannibalism may begin when fish are 14.5 mm long, whereas our experimental results demonstrated that cannibalism may occur at 10–11 mm.

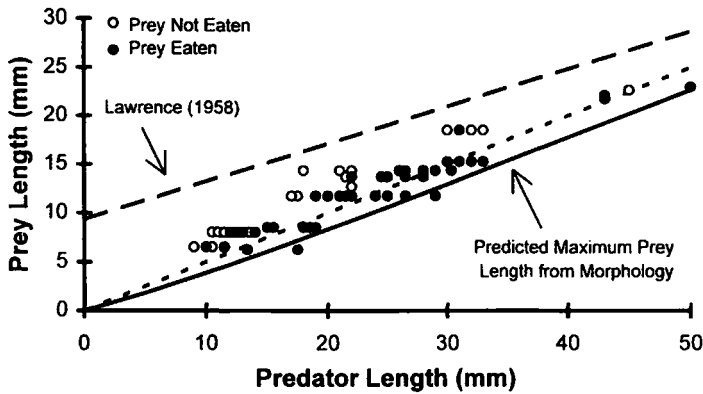


FIGURE 2.—Results of morphological measurements and experimental trials quantifying the maximum prey size an age-0 largemouth bass of given length can ingest. The solid line represents predictions made by our morphological data, the dashed line represents predictions of Lawrence's (1958) morphological data, and the dotted line is a predator-prey ratio of 2:1.

Discussion

The relationship of total length and body depth (Figure 1) was similar to that found by Lawrence (1958). The relationship of total length and mouth width, however, differed from that of Lawrence (1958) when fish were less than 70 mm (Figure 1). The difference may result from extrapolating Lawrence's results to smaller fish or possible differences in measuring techniques. Our experimental results demonstrate that the relation between predator length and maximum prey length derived from morphological measurements underrepresent the ability of 10–50-mm age-0 largemouth bass to consume other age-0 largemouth bass. The results suggest that a predator to prey length ratio of 2:1 may more closely approximate the observed relationship (Figure 2).

The possibility of cannibalism at an early age presents an important alternative prey resource for early spawned individuals. This potential food source may create a window of opportunity for largemouth bass growth because a diet of fish, compared to a diet of invertebrates, increases net energy gain (Timmons et al. 1980; Keast and Eadie 1985). The transition from an invertebrate diet to a fish diet may create major shifts in growth of largemouth bass. Age-0 largemouth bass that switch to piscivory earlier than conspecifics (either because of an initial growth advantage or earlier hatching date) may realize increased growth (Aggus and Elliott 1975; Kitchell et al. 1994). In contrast, late-spawned or slower-growing fish may be subjected to higher mortality as older or faster-growing fish become cannibalistic (DeAngelis et al. 1979; Brabrand 1995).

For intracohort cannibalism to occur, however, there must be a sufficient difference in size between early spawned and later-spawned individuals. This, in turn, depends on the duration of spawning and initial growth rates of age-0 largemouth bass. The spawning duration of largemouth bass in northern lakes such as Paul Lake ranges between 17 and 26 d (Schmidt and Fabrizio 1980; Post, unpublished data), and growth averages 0.76 mm/d (average growth rate for fish less than 25 mm in Paul Lake, 1993–1995; Post, unpublished data). Our morphological analyses indicate that age-0 largemouth bass can become cannibalistic as early as 11 d after swim-up (8 d, considering experimental results). Thus, the duration of spawning is sufficient for intracohort cannibalism to occur, considering only morphological constraints.

However, the nest school guarding behavior of male largemouth bass likely prevents cannibalism until age-0 largemouth bass are abandoned. Male guarding can prevent cannibalism by maintaining spatial segregation between nest schools and through direct nest school protection. Furthermore, lengths are generally too similar within a nest school to allow intra-nest school cannibalism. If we assume that male guarding prevents cannibalism until prey reach 15–20 mm (approximate size of male abandonment; Brown 1985), a spawning period of 25–43 d is required for intracohort cannibalism. This suggests that intracohort cannibalism in northern lakes may be possible only for fast-growing, early spawned individuals preying upon late-spawned, slow-growing individuals.

In three Mississippi reservoirs, Goodgame and Miranda (1993) reported a spawning duration of

60–71 d and average growth rate of 0.71–0.93 mm/d. Our morphological analyses suggest that cannibalism may occur about 9 d after eggs hatch in these reservoirs. Again, if we assume that male guarding prevents cannibalism until prey reach 15–20 mm, a spawning period of 20–32 d is required for intracohort cannibalism to occur. Thus, intracohort cannibalism may be common in these southern reservoirs and may strongly influence largemouth bass recruitment.

Our study focused on the morphological constraints of cannibalism by age-0 largemouth bass. However, other factors such as predator and prey behaviors (including schooling and male guarding), encounter rates, and the availability of alternative fish prey must also be considered to fully understand the ecological importance of this predator–prey relationship. Our results show that intracohort cannibalism can begin at an early age, yet because of male guarding and a short spawning season, intracohort cannibalism may be less significant in northern lakes. In contrast, our analysis suggests that intracohort cannibalism is more likely in southern lakes and reservoirs where higher growth rates and an extended spawning season allow a greater size difference between early and late-spawned individuals.

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